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FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25348

FMI Title: GDAS Digital Power Supply Replacement

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

The following CT Equipment is affected:

Number of Systems	Gantry S/N	Digital PS	Customer Order #	Location	Customer Name
1	340619CN6	NB2728564	2520197	Japan	KYOUWAIKI KABUSHIKIGAISSYA
2	340620CN4	NB2728535	2005016	Port Orange, FL	KEM ASSOCIATES
3	341854CN8	NB2728564	2539991	Oconomowoc, WI	OCONOMOWOC MEM HOSPITAL
4	345041CN8	NB2728519	2537843	Israel	M.A.R. LTD.
5	346215CN7	NB2728530	2531841	Thailand	BUMRUNGRAD MEDICAL CENTER LTD
6	346337CN9	NB2728559	2537544	Portland, OR	PROVIDENCE HOME SERVICES
7	346495CN5	NB2728562	2520658	Canada	ST JOSEPHS HOSPITAL
8	346616CN6	NB2728561	2533551	France	HOPITAL LA TIMONE
9	346712CN3	NB2728568	2530865	Phoenix, AZ	ST JOSEPHS HOSP AND MED CENTER
10	347350CN1	NB2728567	2536137	Spain	ERESA
11	347514CN2	NB2728565	18969	Poland Advanced order	No customer name available
12	349669CN2	NB2728572	2535955	India	GMR INSTITUTE OF IMAGING & RESEARCH MRI SCAN CTR
13	348138CN9	NB2728566	2537460	Finland	OULUN YLIOPISTOLLINEN SAIRAALA
14	348144CN7	NB2728544	2536334	Singapore	CHINESE GENERAL HOSPITAL AND MEDICAL CENTER
15	348289CN0	NB2728541	2537706	Norway	HELGELANDSYKEHUSET HF, AVD. SANDNESSJØEN
16	348625CN5	NB2728537	2529212	Germany	PROF. MED. STEINER, DR. MED. GUST, DR. MED. AUFFERMANN
17	348624CN8	NB2728557	2537734	Hungary	EUROMEDIC DIAGNOSTICS BV
18	348953CN1	NB2728558	2538109	Taiwan	TZU-CHI BUDDHIST GENERAL HOSP

Number of Systems	Gantry S/N	Digital PS	Customer Order #	Location	Customer Name
19	349338CN4	NB2728569	2539769	UK	GOLDEN JUBILEE NATIONAL HOSPITAL
20	350418CN0	NB2728545	2530231	Bristol, TN	MINNESOTA MEDICAL IMAGING LLC
21	35043CN1	NB2728574	2538202	Boston, MA	MASSACHUSETTS GENERAL HOSPITAL
22	350477CN6	NB2728571	2538204	Boston, MA	MASSACHUSETTS GENERAL HOSPITAL
23	35040CN1	NB2728577	2538569	Cincinnati, OH	UNIV HOSPITAL

Purpose

The purpose of this FMI is to change the GDAS Digital power supply chassis with a supplied new one due to an internal cable issue that could result in an Image Quality issue. This issue is a manufacturing defect on a cable shield by the cable manufacturer that is limited to a specific subset of Digital power supplies only.

Prerequisite FMIs

None

Related FMIs

None

Furnished Materials

Qty	Description	Part #
1	FMI Document	5121148-100
1	GDAS Digital Power Supply	2330352-3
2	Power supply return label	N/A
1	DVD (for scan file return)	2337414
1	GDAS center backplane label	N/A
1	DEFECTIVE PART - LABEL	2358146

Tools Required

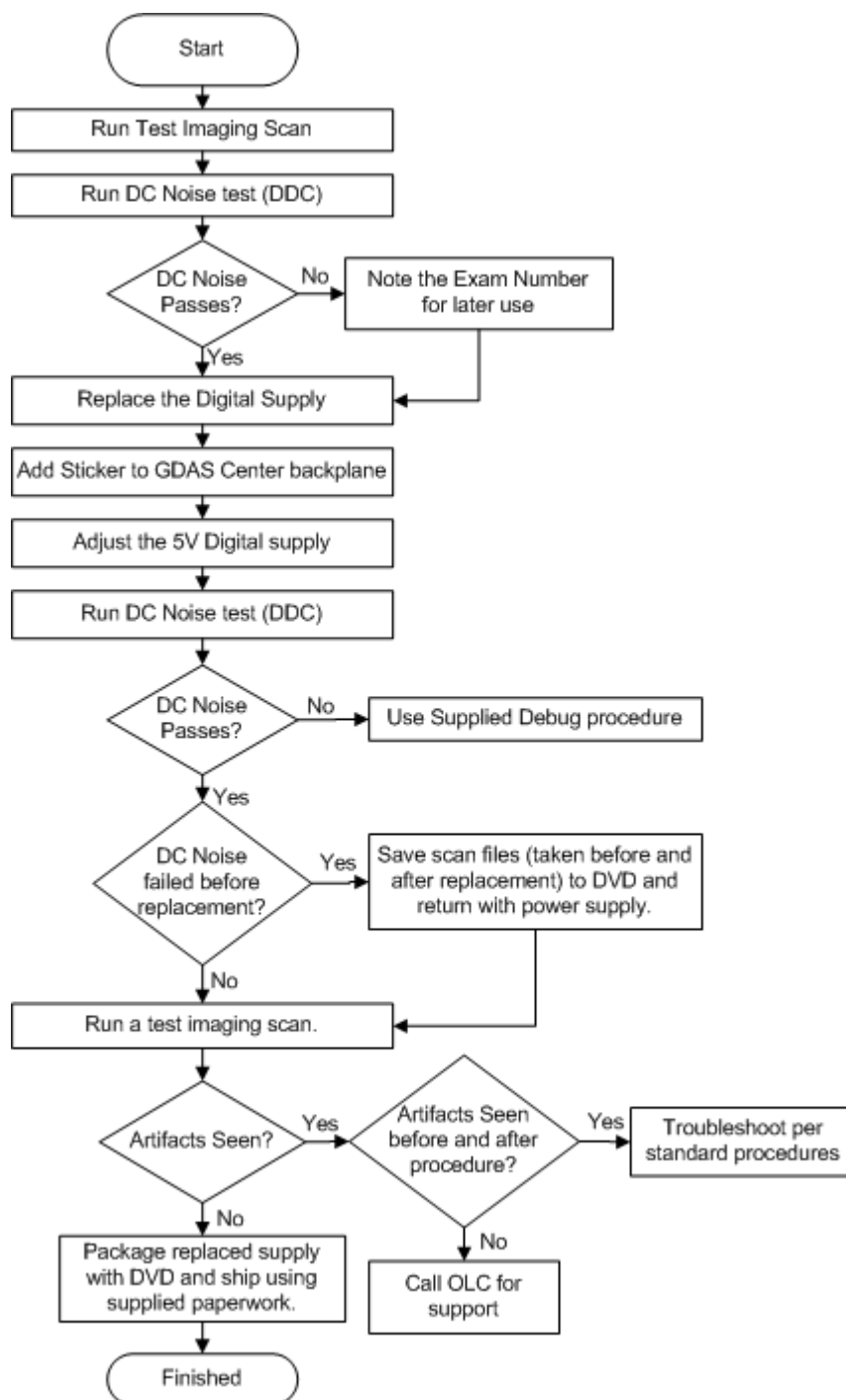
Qty	Description	Part #
1	FMI Document	5121148-100
1	10mm long hex driver	Supplied with system (used during tube changes)
1	High capacity Torque Wrench for H-power gantry balance	Order from Tool Depot prior to site visit. 46-315668P3
1	Field Torque wrench kit	Contact local zone support if you don't have one.

People Power

3 hours labor and 1.5 hours of travel

Procedure

FMI Overview



Preparation

1. Using the water phantom, perform a generic service scan (service protocol 42.1) to check for existing image artifacts. If an artifact exists continue with this FMI to see if the artifact is related to the Digital power supply cable defect before troubleshooting further. See Figure 1 for a sample an artifact that could be caused by the power supply cable problem.

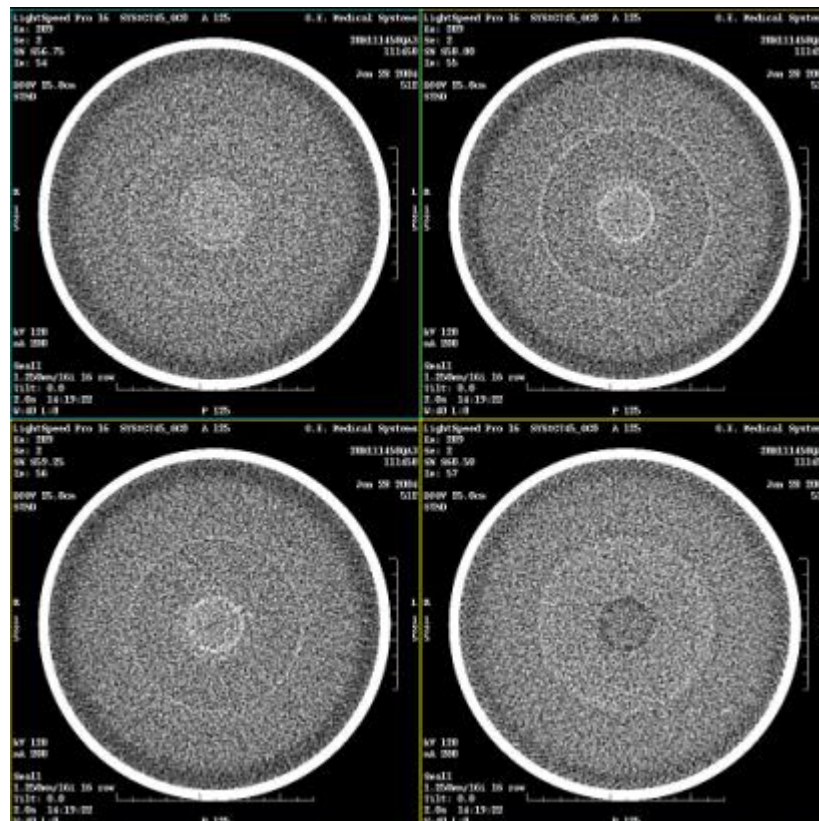


Figure 1 Sample images showing power supply related issue

2. End the exam and start the DAS Tools diagnostic from the Common Service Desktop, Diagnostics menu.
3. Select **[Manual Test]**, and select 1 iteration of **DC Noise & DC Offsets** test selection. Set other test selections to zero iterations.
4. Select **[Accept]** to run the test.
5. When the test completes check the GUI to see if **Test Successes** is "1" indicating a pass. Test Failures will be shown as "1" for a failed test.
 - If the DC Noise test passes continue with the Digital power supply replacement in the next section.
 - If the DC Noise test fails perform the following steps:
 - a) Dismiss the DAS Tools GUI.

- b) Start the DDC diagnostic.
- c) Select a slice collimation of 16x0.625, Static Xray off, Set DAS Gain to 3. Leave all other parameters as default.
- d) Select **[Accept Rx]** and hit the **[Start Scan]** hard key when indicated to run the scan.
- e) Start **[Scan Analysis]** from the Common Service Desktop, Image quality menu.
- f) Select Update to make sure all exams are shown. Document in the table below the exam number just completed that matches the date and time of the DDC scan.

Table 1 DDC Exam Number

Exam Number:	
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- g) Dismiss Scan analysis and continue with the power supply replacement as shown in the next section.

Digital power supply replacement Procedure

1. Position the table to its lowest position, fully out of the gantry.
2. Remove the gantry right and left side covers.
3. Remove the gantry top covers.



NOTICE

Always turn OFF the HVDC before the 120VAC. Turning OFF 120VAC power before HVDC power can result in equipment damage.



NOTICE

Use proper lockout/tagout procedures before replacing components.

4. Turn OFF the Axial Enable and HVDC switches on the service switch panel.
5. Rotate the gantry to access the Digital power supply on the left side (same side as TGP) then turn off the 120VAC service switch.



WARNING

CRUSH HAZARD.

UNEXPECTED GANTRY ROTATION CAN CAUSE SERIOUS INJURY.

NEVER SERVICE THE GANTRY WITH THE AXIAL DRIVE ENABLED.

6. Lock the gantry rotation.

7. Disconnect the harness over the Digital supply by loosening the captive screws of the harness brackets from the DAS back plate.
8. Disconnect the cables from the supply and neighboring components as necessary. See the following list for items to pay attention to.
 - o Check on each end of the power supply for cables that may be mounted just out of sight on the ends. Most are secured using reusable cable clamps that are black and shown in [Figure 2 GDAS Cabling](#). To open the cable clamps pull back on the small tab away from the end of the strap and push the strap back through the clamp.
 - o There are cables on the end of the Digital power supply that must be removed. The suggested order of removal is: DMB plug, Fan Plug, DHCB plug, Collimator plug. These cables have connectors with tabs that also must be removed from the cable interface plate to remove the supply. To remove the Molex connectors from the bulkhead after disconnecting the cable use a small flat screw driver to push down on the tab and pull the Molex back to release the top then reach under the Molex with the screw driver and push up on the bottom tab while pulling on the connector. It should pop out with both tabs released from the bulkhead.
 - o The Detector heater wires are small and may break with repeated bending. Use caution with this cable.
 - o Black cable clamps are reusable ties. Do not cut them.

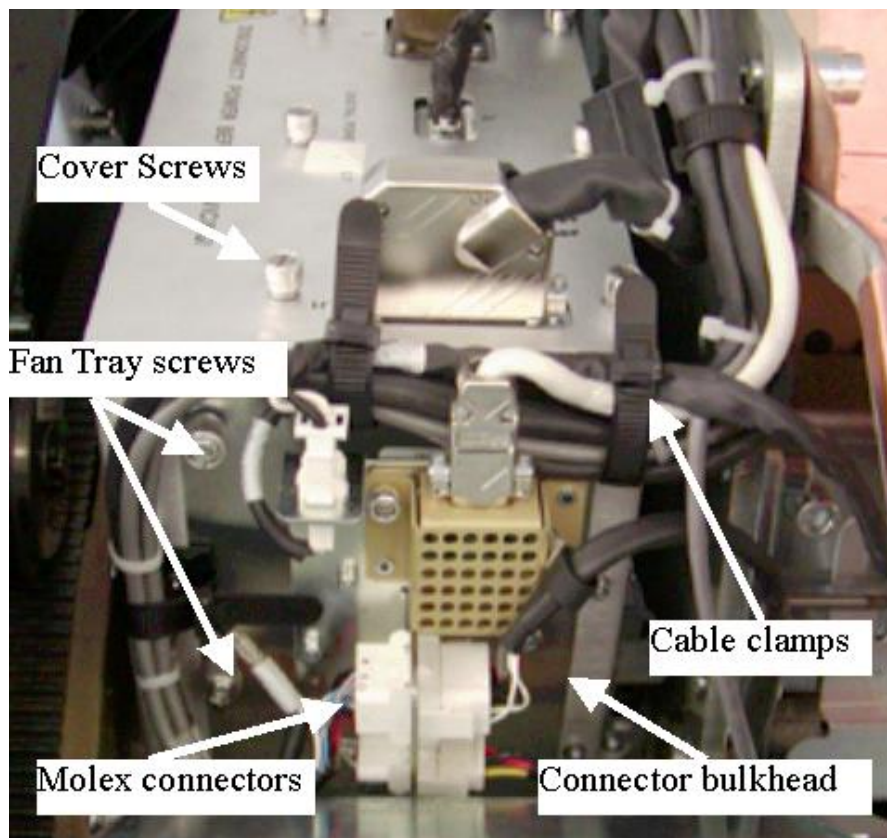


Figure 2 GDAS Cabling

9. Move the harness out of the way by gently pulling it up and over the DAS plate.

For the Digital supply, the cable from J2 on the Jedi inverter to J32 on the DAS right chassis backplane is not part of the main cable bundle but still tie-wrapped at a couple points. These tie-wraps may need to be disconnected to provide slack for the harness to be moved up and over the DAS plate.

CAUTION



Burn injury possible.

Power supply may be hot to touch.

Use caution to prevent possible burn injury from hot surface contact.

10. Remove the top cover of the power supply. Loosen the 3 captive screws holding the fan tray assembly (1 on top and 2 on the side). These are all combination heads of Torx and medium straight screwdriver. The Fan tray bottom screw on the side of the Digital power supply can be hard to get to. A 4mm hex wrench (non-ball end) will fit this T30 Torx head to allow easier removal and tightening. Disconnect the Fan tray power plug and pull out the fan tray.

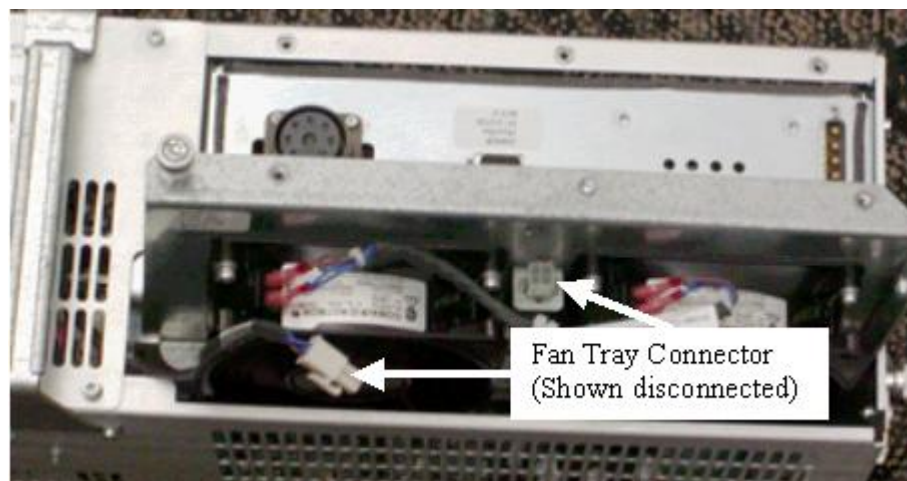


Figure 3 Power supply showing fan tray



Figure 4 Power supply with fan tray removed

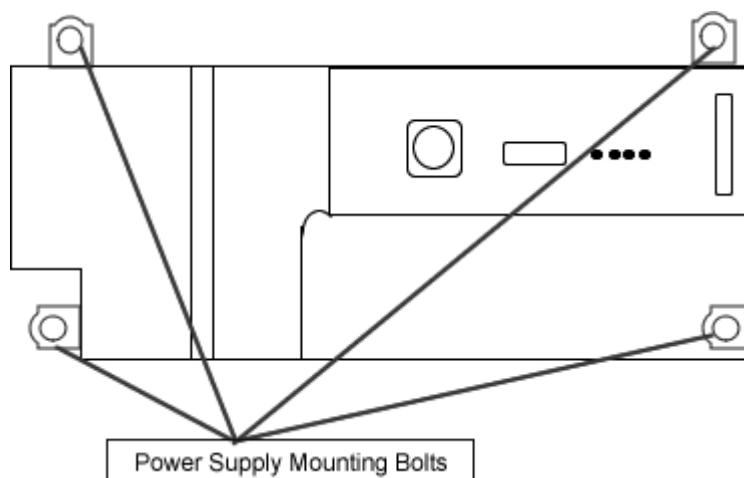


Figure 5 Power supply mounting bolt locations

NOTE: Be sure all cables are disconnected before removing power supply.

NOTE: Be careful not to hit the home flag sensor board when removing supplies on the TGP side of the gantry.

11. Remove the four bolts shown in Figure 5 Power supply mounting bolt locations that are used to mount the power supply to the rotating base. Remove the power supply assembly from the gantry by grabbing the handle in one hand and the lower side of supply in other, to steady it. The cable bulkhead on the side of the Digital supply can be used as a second handle.
12. The cable bulkhead on the side of the digital power supply will need to be transferred from the old supply to the new one.

**NOTICE**

Failure to follow the process defined below may cause a mounting failure.

13. Insert the new digital power supply chassis reversing the steps above. Use the following process to Torque the power supply bolts:
14. First Torque the bolts to 50% of the final value using the following settings.

Table 2 Initial Torque setting

33.2 N-m	25 lb-ft	294 lb-in	338 kg-cm
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15. After all 4 bolts have been seated using the initial torque setting apply the final torque as shown in the following table.

Table 3 Final Torque values

66.4 N-m	49 lb-ft	588 lb-in	677 kg-cm
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**NOTICE**

The connector thumbscrews on the GDAS power supply and backplane connector harnesses **WILL BREAK** very easily if using a large diameter handle screwdriver. A gentle hand torque using a small diameter handle screwdriver is recommended.

16. Reconnect the cabling harness. Make sure all cable clamps are secure and any tie-wraps removed are reinstalled.

17. Install supplied label on the DAS center backplane in the location shown in [Figure 6 Center Backplane sticker and location](#). This sticker is a reminder that the 12V test point is not used on GDAS systems. This backplane will be common to MDAS16 and GDAS.

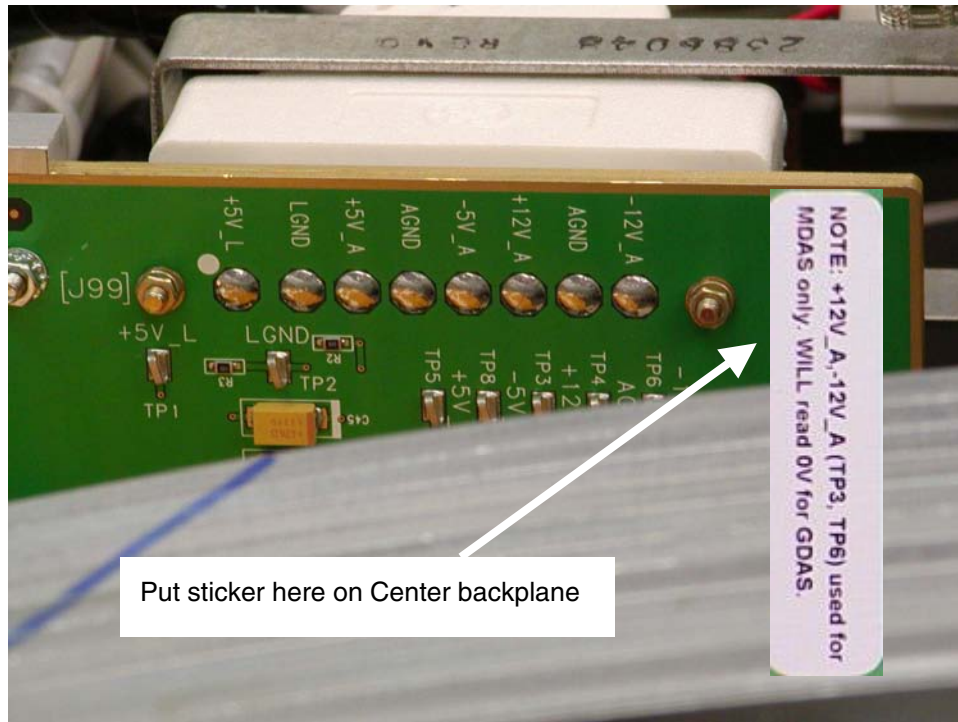


Figure 6 Center Backplane sticker and location

18. Restore power to the gantry. You may need to run a **Scan Reset** from the **Scan Resets** selection on the Common Service Desktop.
19. From the Common Service Desktop at the console start the DAS Tools diagnostic and run the **[Aux Channels]** test to read the power supply voltages at the DCB. Click **[View Log]** and find today's date to see the readout of the power supplies. Use the voltage levels reported by this tool, shown by selecting **[View Log]**, for adjustment to the specifications in [Table 4 Power Supply Levels](#). Only the 5V digital supply may need to be adjusted.

Table 4 Power Supply Levels

Location	Test Point	Description	DC Level Specification	Noise and Ripple p-p specification
DAS center backplane	TP1	+5V Digital w.r.t LGND	+5V + 0.05Vdc	5mv in 20Mhz Bandwidth
DAS center backplane	TP2	Logic Ground (LGND)		

20. After looking at the Aux Channel report, you can connect a DVM to the test points labeled in Table 3 for a local reference when adjusting the pot. Note the difference between the test point value and the Aux Channel report and adjust accordingly.
21. Continue to adjust and rerun Aux Channel test until the Aux channel report for the supply matches specifications.
22. When done adjusting the power supply level, start the DAS Tools diagnostic from the Common Service Desktop Diagnostics menu.
23. Select **[Manual Test]**, and select 1 iteration of **DC Noise & DC Offsets** test selection. Set other test selections to zero iterations.
24. Select **[Accept]** to run the test.
25. When the test completes check the GUI to see if **Test Successes** is "1" indicating a pass. Test Failures will be shown as "1" for a failed test.
 - If the DC Noise test failed prior to power supply replacement and passes now, save the DDC scanfile identified in Table 2 (Preliminary steps section) onto the provided DVD using the instructions below. Place the DVD in the power supply box to be returned with the supply. Go to the Finalization section.
 - a) Place the supplied DVD into the console DVD drive.
 - b) Click on **[RECON MGMT]** icon on the scan desktop
 - c) Click on **[Save Scan Data]**
 - d) Select the Exam number identified in Table 2 (Preliminary steps section), then select the series and scan for that Exam.
 - e) Select **[Save Selected Scanfiles]**
 - If the DC Noise test passed both before and after power supply replacement go to the Finalization section.
 - If the DC Noise test fails now save both DC Noise scan files (run before and after power supply replacement) to DVD per the instructions above. Perform the Debug procedure below.

Debug Procedure (only if supply fails DC noise after replacement)

Perform the following checks to help isolate a potential failure mode.

Power Supply Levels:

1. From the Common Service Desktop start the DAS Tools diagnostic and run the Aux Channels test to read the power supply voltages at the DCB. Check all DAS power supply voltages (except +/-12V) on Aux channels. Reference [Table 5 GDAS Power supply voltages](#). Use the test points as a local reference when adjusting the supply only. Use the Aux Channel output as comparison against specifications.

Table 5 GDAS Power supply voltages

Location	Test Point	Description	DC Level Specification (Seen by Aux Channels)	Noise and Ripple p-p specification
DAS center backplane	TP1	+5V Digital w.r.t LGND	+5V + 0.05Vdc	5mv in 20Mhz Bandwidth
DAS center backplane	TP5	+5V Analog w.r.t AGND	+5V + 0.05Vdc	5mv in 20Mhz Bandwidth
DAS center backplane	TP8	-5V Analog w.r.t AGND	-5V + 0.05Vdc	5mv in 20Mhz Bandwidth
DAS center backplane	TP2	Logic Ground (LGND)		
DAS center backplane	TP7	Analog Ground (AGND)		

2. If any power supply levels required adjustment, return to Step 22 of the main procedure to rerun the DC Noise test otherwise continue with the next check.
3. Exit the DAS Tools GUI.

60Hz Noise check:

4. Start the DDC diagnostic.
5. Select a slice collimation of 16x0.625, Static X-ray off, Set DAS Gain to 3. Leave all other parameters as default.
6. Select **[Accept Rx]** and run the scan.
7. From the Common Service Desktop start **Scan Analysis**; select the scan file matching the date/time of the DDC scan. Select **[Plot VVC]** to view the data from the DDC scan. Select **"FFT Channel Data"** option. Figure 7 below shows a good noise data sample and Figure 8 shows a bad noise data sample.
8. If the data plot seen matches Figure 8, continue with the Backplane resistor check and save this scan file to the supplied DVD along with the scan file taken prior to power supply replacement. If the data plot matches Figure 6, call the OLC for support.

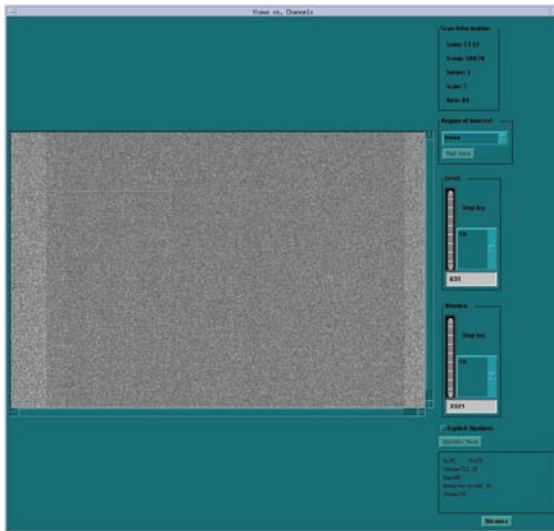


Figure 7 Noise data - good sample

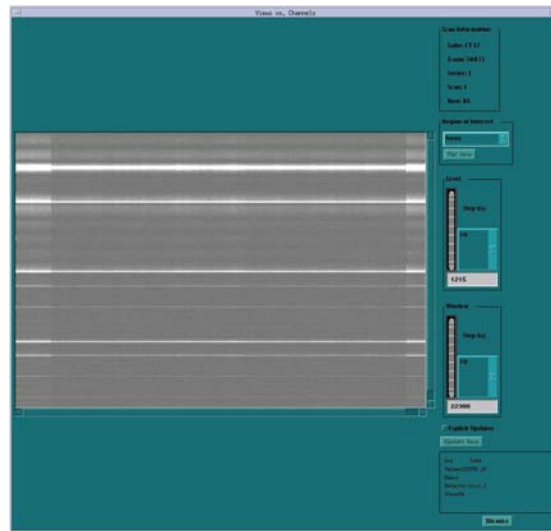


Figure 8 Noise data - bad sample

Backplane Resistor Check:

9. Perform a visual inspection of the 0-ohm resistors R2, R3. See [Figure 9 DAS Center Backplane behind 180 weight stack](#).

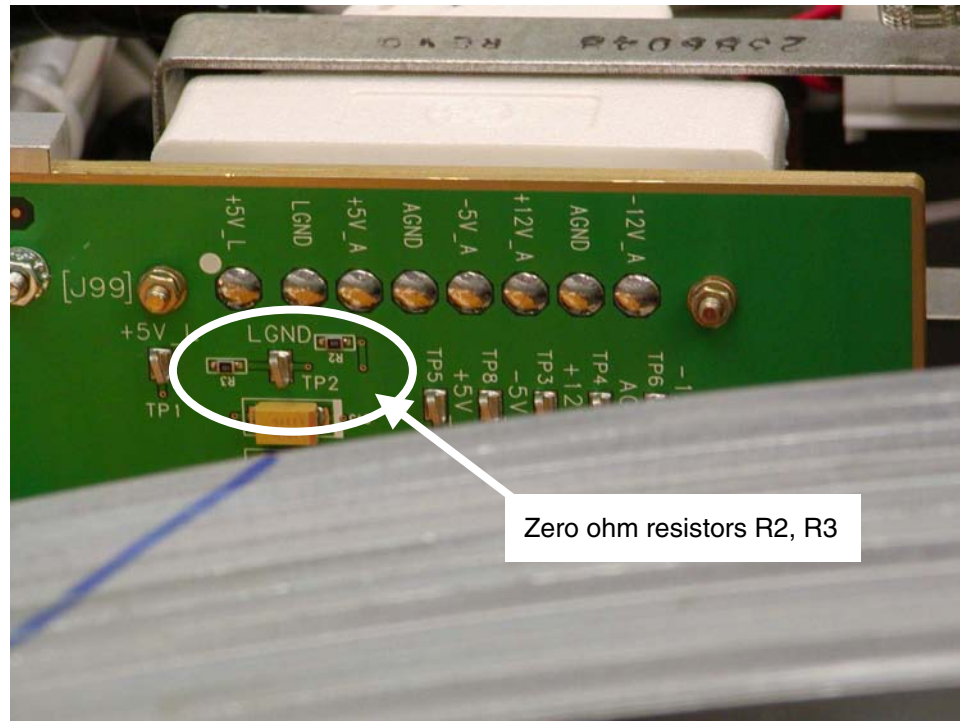


Figure 9 DAS Center Backplane behind 180 weight stack

If there is no visual damage perform the following steps:

- a. Unplug the Digital and Analog Power Connectors from Center Backplane J104 (Digital) and J96 (Analog).
- b. Test 1: Using a meter, check resistance across resistors R2, R3. Should be less than 1 ohm across each resistor.
- c. Test 2: Using a meter, on Power connectors of unplugged cable (still connected to Power Supply), check that chassis ground is NOT connected to AGND/LGND Pin 2 to cable ground shell, pin 4 to cable ground shell, and pin 7 to cable ground shell. Should be greater than 100K ohms.

Debug Results

- d. If Test 1 fails – Order and Replace the Center Backplane Part # 2388723.
- e. If Test 2 fails, Order and Replace the GDAS Power Supply Part # **2330352-3** and the Center Backplane Part # 2388723. Place label on the power supply and ship the supply back using the supplied return paperwork.

Call OLC for support if needed.

Finalization

1. From the **Common Service Desktop, Calibration menu** select [**Gantry Service balance**] and perform a balance check. Rebalance the gantry if necessary.
2. Using the water phantom, perform a generic service scan to make sure system scans without artifacts.
 - a. If no artifacts are seen, continue with step 3.
 - b. If an artifact is seen before and after this FMI procedure, troubleshoot per normal image artifact troubleshooting procedures. Continue with step 3.
 - c. If no artifact is seen before this FMI and an artifact appears after this procedure: Call OLC for help.
3. Place the removed Digital power supply in the same box as the delivered supply. Place sticker on the power supply identifying it as being returned per this FMI.
4. Place the DVD containing saved scan data files in the box with the power supply.
- 5. Using the supplied return paperwork, place the return label on the box and ship the removed power supply and DVD.**
- 6. Close this FMI.**